

Final Portfolio Project
Option 2: Encoder-Decoder Model for Sequence-to-Sequence Prediction

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Introduction

Deep learning models are often discussed in terms of their ability to classify information or predict a single outcome. However, many real-world problems require a model to transform one sequence of information into another sequence. Encoder-decoder models were developed to address this type of sequence-to-sequence problem. In a basic encoder-decoder architecture, the encoder processes an input sequence and creates an internal representation of that information. The decoder then uses the encoded representation to generate an output sequence. Long short-term memory (LSTM) networks are particularly useful within this architecture because they are designed to retain information across sequential inputs. Sutskever et al. (2014) demonstrated this approach using one LSTM to encode an input sequence and another LSTM to generate the corresponding output sequence. Their work showed that sequence-to-sequence learning could successfully address complex problems such as language translation without requiring the input and output sequences to have identical structures or lengths.

The value of this architecture extends beyond language processing. Many industries generate data in which order, history, and changes over time contain important information. Healthcare records contain sequences of encounters and treatments, transportation systems must reason about movement over time, and energy systems use historical demand to forecast future demand. In each case, an encoder-decoder architecture can learn a representation of what has already occurred and use that representation to generate a meaningful future sequence. The following use cases demonstrate how this general architecture can be applied to distinctly different problems.

Healthcare and Longitudinal Clinical Prediction

Healthcare provides a particularly compelling application for encoder-decoder models because a patient's medical history is inherently longitudinal. Diagnoses, medications, laboratory results, procedures, and clinical encounters occur over time, and the significance of a new event may depend on what occurred during previous visits. Rather than treating each encounter as an isolated data point, sequence-based models can represent the patient's prior history as an ordered input and use that information to predict a future clinical sequence.

A strong example is TransformEHR, an encoder-decoder model developed by Yang et al. (2023) for predicting disease outcomes from longitudinal electronic health records (EHRs). The researchers pretrained the model using records from approximately 6.5 million patients. The encoder processed information from previous patient visits, while the decoder generated the diagnostic codes associated with a future visit. Importantly, the model incorporated visit dates rather than relying solely on the order in which events appeared in the record, allowing temporal information to become part of the representation of the patient's history.

This application illustrates an important advantage of encoder-decoder models in healthcare: clinical outcomes rarely occur independently. A future visit may involve multiple related diagnoses, and those outcomes may be influenced by combinations of conditions documented throughout the patient's history. TransformEHR therefore generated the diagnostic codes for the next visit sequentially, using previously generated codes along with information encoded from earlier visits. The researchers reported that the model outperformed prior approaches on multiple

prediction tasks, including improvements in area under the precision-recall curve for pancreatic cancer onset and intentional self-harm among patients with post-traumatic stress disorder.

The potential benefit extends beyond simply predicting a diagnosis. An encoder-decoder architecture provides a framework for reasoning about a patient's trajectory rather than a single snapshot in time. In complex medical histories, patterns may emerge across multiple encounters that are difficult to recognize when records are evaluated independently. At the same time, prediction should not be confused with clinical judgment. An encoder-decoder model can identify patterns and potential future outcomes, but its predictions remain dependent on the completeness and quality of the underlying EHR data. For this reason, these systems may be most valuable as clinical decision-support tools that surface longitudinal patterns for clinicians rather than independently determining diagnoses or treatment.

Transportation and Trajectory Prediction

Transportation is another area in which sequential prediction is particularly important because movement is defined by what occurs over time. An autonomous vehicle, for example, cannot safely respond to a pedestrian based only on the pedestrian's current location. It must consider the person's previous positions and direction of travel to estimate where that individual is likely to move next. In this sense, trajectory prediction can be treated as a sequence-generation problem: a historical sequence of positions provides the input, while a predicted sequence of future positions becomes the output. Alahi et al. (2016) explored this problem using an LSTM-based model to predict pedestrian trajectories in crowded environments.

One challenge with trajectory prediction is that an individual's movement does not occur independently. Pedestrians adjust their paths in response to other people by yielding, maintaining personal space, traveling together, and avoiding collisions. Alahi et al. (2016) developed Social LSTM, which allowed multiple LSTMs representing nearby pedestrians to share information through a pooling layer. This allowed the model to learn interactions between neighboring trajectories rather than relying exclusively on manually defined rules about how people should behave. The researchers reported that their approach outperformed previous trajectory-forecasting methods by more than 42% across their evaluation.

The benefit of sequence-based models in transportation is therefore more significant than simply predicting a person's next location. The model can use a sequence of past observations to generate a sequence representing how movement may continue, while also accounting for interactions that influence that movement. This capability has applications in autonomous vehicles, socially aware robotics, pedestrian monitoring, and other intelligent transportation systems where anticipating future motion can contribute to safer decision-making. At the same time, trajectory predictions represent probable behavior rather than certainty. Unexpected actions, environmental changes, or situations that differ substantially from the model's training data can still produce incorrect predictions. In safety-critical transportation systems, these predictions should therefore contribute to a broader decision-making system rather than function as the sole basis for an autonomous action.

Energy and Multistep Demand Forecasting

Energy and utility systems provide another useful application of encoder-decoder models because electricity demand changes continuously over time. Utilities must anticipate not only how much electricity will be required at a single future moment, but how demand may change across several future time intervals. Historical consumption, weather conditions, time of day, day of the week, and seasonal patterns can all influence future demand. This makes energy forecasting a multistep sequence prediction problem in which a sequence of historical observations can be used to generate a sequence of future values.

LSTM-based encoder-decoder architectures are well suited to this type of forecasting because the encoder can learn patterns contained within historical observations while the decoder generates predictions over multiple future time steps. Marino et al. (2016), for example, investigated deep neural networks, including LSTM models, for predicting energy consumption in residential buildings. Their research demonstrated the value of recurrent neural networks for modeling temporal dependencies in energy data and showed that LSTM-based approaches could outperform conventional forecasting approaches under several experimental conditions.

More specifically, sequence-to-sequence architectures can support multi-horizon forecasting, where the goal is to predict an entire future interval rather than repeatedly make isolated one-step predictions. This is valuable to utilities because energy planning is inherently forward-looking. Forecasting the expected pattern of demand across the next several hours or days can support decisions about electricity generation, grid balancing, energy storage, and the integration of renewable resources. Encoder-decoder models also allow the length and structure of the historical input window to differ from the future forecasting window, which provides greater flexibility than models designed to produce only a single output.

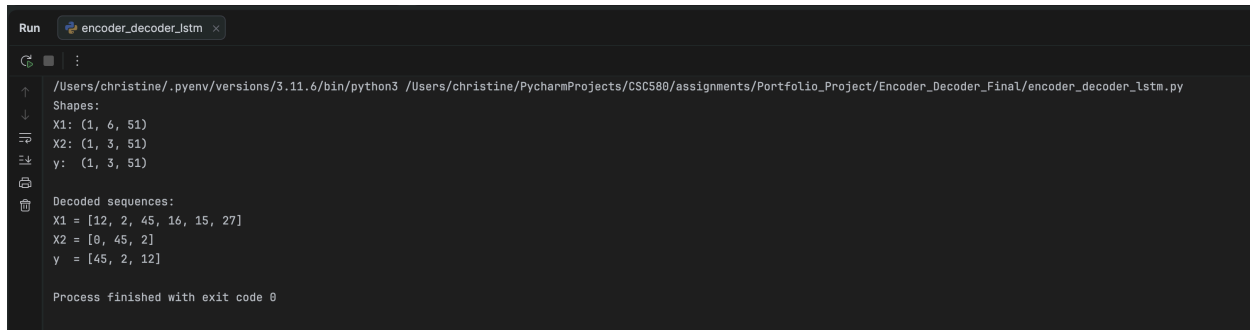
The benefit of encoder-decoder models in this industry is therefore their ability to transform historical temporal patterns into a sequence of actionable future estimates. However, as with healthcare and transportation, these predictions remain dependent on the data used to generate them. Extreme weather, sudden changes in consumer behavior, equipment failures, or other unusual events may differ from historical patterns and reduce forecast accuracy. For utilities, an encoder-decoder forecast is therefore most useful as one component of a larger planning and decision-support process rather than as a guarantee of future demand.

Encoder-Decoder LSTM Implementation and Results

To demonstrate sequence-to-sequence learning, an encoder-decoder LSTM was developed in Python using TensorFlow and Keras. The model was given sequences of six randomly generated integers and trained to return the first three integers in reverse order. As illustrated in **Figure 1**, the source sequence, shifted decoder input, and target sequence were one-hot encoded using 51 possible values, with zero reserved as the decoder start token. The encoder and decoder each used 128 LSTM units, followed by a dense softmax layer that generated the predicted output sequence.

Figure 1

Sequence-to-Sequence Dataset Generation and One-Hot Encoding Validation

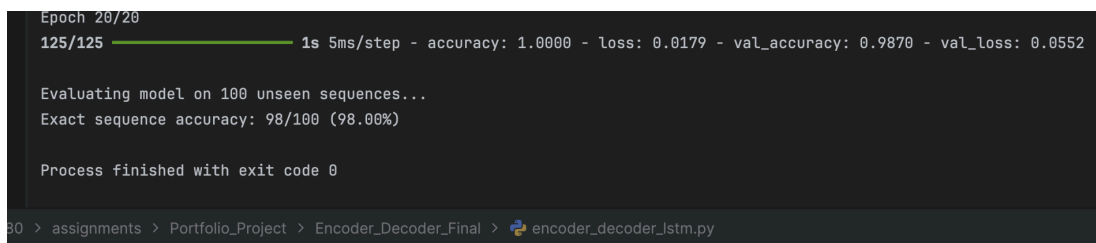


```
Run encoder_decoder_lstm x
/Users/christine/.pyenv/versions/3.11.6/bin/python3 /Users/christine/PycharmProjects/CSC580/assignments/Portfolio_Project/Encoder_Decoder_Final/encoder_decoder_lstm.py
Shapes:
X1: (1, 6, 51)
X2: (1, 3, 51)
y: (1, 3, 51)
Decoded sequences:
X1 = [12, 2, 45, 16, 15, 27]
X2 = [0, 45, 2]
y = [45, 2, 12]
Process finished with exit code 0
```

The overall training and inference process is summarized in Figure 2. During training, the encoder processed the complete source sequence and passed its hidden and cell states to the decoder. The decoder learned to generate the target sequence using a shifted version of the expected output. During inference, the encoder processed an unseen source sequence once, after which the decoder generated predictions recursively, passing its state and previous output into the next prediction step.

Figure 2

Encoder-Decoder LSTM Training and Exact-Sequence Evaluation Results



```
Epoch 20/20
125/125 ————— 1s 5ms/step - accuracy: 1.0000 - loss: 0.0179 - val_accuracy: 0.9870 - val_loss: 0.0552
Evaluating model on 100 unseen sequences...
Exact sequence accuracy: 98/100 (98.00%)
Process finished with exit code 0
30 > assignments > Portfolio_Project > Encoder_Decoder_Final > encoder_decoder_lstm.py
```

After 20 epochs, the model achieved approximately 99% validation accuracy. Exact-sequence testing provided a stricter evaluation because all three predicted values had to match the target sequence for the prediction to be counted as correct. Across independent runs, the model correctly predicted approximately 96–98% of 100 unseen sequences. As shown in Figure 3, the model also successfully generated complete target sequences for individual test examples.

Figure 3

Encoder-Decoder LSTM Exact-Sequence Evaluation and Sample Predictions

```
Evaluating model on 100 unseen sequences...
Exact sequence accuracy: 96/100 (96.00%)

Sample predictions:
X=[2, 20, 17, 5, 24, 31] y=[17, 20, 2] yhat=[17, 20, 2]
X=[40, 7, 49, 7, 38, 32] y=[49, 7, 40] yhat=[49, 7, 40]
X=[49, 22, 40, 40, 44, 17] y=[40, 22, 49] yhat=[40, 22, 49]
X=[41, 3, 30, 29, 43, 24] y=[30, 3, 41] yhat=[30, 3, 41]
X=[44, 17, 5, 8, 22, 30] y=[5, 17, 44] yhat=[5, 17, 44]
X=[13, 6, 49, 47, 34, 26] y=[49, 6, 13] yhat=[49, 6, 13]
X=[50, 44, 46, 34, 27, 14] y=[46, 44, 50] yhat=[46, 44, 50]
X=[28, 1, 46, 50, 36, 46] y=[46, 1, 28] yhat=[46, 1, 28]
X=[43, 27, 17, 10, 38, 43] y=[17, 27, 43] yhat=[17, 27, 43]
X=[40, 11, 39, 9, 29, 48] y=[39, 11, 40] yhat=[39, 11, 40]

Runtime predictions saved to: runtime_predictions.txt

Process finished with exit code 0

> assignments > Portfolio_Project > Encoder_Decoder_Final > encoder_decoder_lstm.py
```

The difference between categorical validation accuracy and exact-sequence accuracy illustrates an important consideration when evaluating sequence models. A model may correctly predict most individual elements while still making an error somewhere within a complete sequence. Exact-sequence evaluation therefore provides a more demanding measure of whether the encoder-decoder model learned the intended transformation.

Conclusion

Encoder-decoder architectures provide a flexible approach to problems in which both inputs and outputs contain sequential information. Applications in machine translation, healthcare, transportation, and energy forecasting demonstrate that the architecture can be adapted to problems involving language, clinical histories, physical movement, and temporal demand. The LSTM implementation developed for this project demonstrated the same underlying principle on a controlled sequence-reversal problem, achieving high accuracy on previously unseen data. Although real-world sequence problems are substantially more complex, the experiment demonstrates how an encoder can represent information from a historical sequence while a decoder uses that representation to generate a new sequence. More advanced approaches, including attention mechanisms and transformer architectures, extend this concept by improving how models access and prioritize information across longer and more complex sequences.

References

- Alahi, A., Goel, K., Ramanathan, V., Robicquet, A., Fei-Fei, L., & Savarese, S. (2016). Social LSTM: Human trajectory prediction in crowded spaces. *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 961–971. <https://doi.org/10.1109/CVPR.2016.110>
- Bahdanau, D., Cho, K., & Bengio, Y. (2015). Neural machine translation by jointly learning to align and translate. *International Conference on Learning Representations*.
- Marino, D. L., Amarasinghe, K., & Manic, M. (2016). Building energy load forecasting using deep neural networks. *IECON 2016—42nd Annual Conference of the IEEE Industrial Electronics Society*, 7046–7051. <https://doi.org/10.1109/IECON.2016.7793413>
- Sutskever, I., Vinyals, O., & Le, Q. V. (2014). Sequence to sequence learning with neural networks. *Advances in Neural Information Processing Systems*, 27.
- Yang, Z., Mitra, A., Liu, W., Berlowitz, D., & Yu, H. (2023). TransformEHR: Transformer-based encoder-decoder generative model to enhance prediction of disease outcomes using electronic health records. *Nature Communications*, 14, 7857. <https://doi.org/10.1038/s41467-023-43715-z>